

CLIMATE ASSESSMENT REPORT

KISIP Tool B: Risk & Vulnerability

Context

County: Mombasa
Settlement: Muoroto Simitini
Assessor: Felix Mutua
Assessment Date: 3/23/2026
Status: draft

Dimension Scores (1-3)

Hazard	3.00
Exposure	2.00
Sensitivity	2.00
Adaptive Capacity	2.67

Overall Scores

Vulnerability	-0.67 (Low)
Risk	3.78 (Medium)

Planning Recommendations

Livelihoods

Food Security

- Prioritize food security in all land use planning decisions
- Plan for emergency food reserves and distribution systems
- Develop comprehensive food security protection plans
- Plan for food production areas protected from climate hazards

Grazing/Farm Land

- Monitor grazing land conditions and availability
- Maintain current grazing land use planning

Crops

- Monitor crop production and climate impacts
- Maintain current crop production planning

Livestock

- Monitor livestock health and climate impacts
- Maintain current livestock management planning

Fishstock

- Monitor fish stock levels and water quality
- Maintain current fisheries management planning

Local Commerce

- Monitor business resilience to climate impacts
- Maintain current commercial area planning

Gender

- Monitor gender impacts of climate change
- Maintain gender-responsive planning processes

Conviviality

- Monitor community peace and social cohesion
- Maintain conflict-sensitive planning processes

Education

- Monitor education continuity during climate events
- Maintain current school infrastructure planning

Water For Irrigation / Water For Domestic Use

- Monitor water availability for irrigation and household use
- Maintain current water infrastructure planning

Access to Finance / Income

- Monitor household financial needs related to climate
- Maintain current financial support planning

Disaster Events

- Monitor disaster risks and preparedness
- Maintain current disaster preparedness planning

Health & Safety

Borne Diseases

- Monitor disease patterns and climate impacts
- Maintain current health surveillance planning

Safety, Location, Urban Planning & Conditions

- Monitor housing location safety and climate impacts
- Maintain current housing safety planning

Solid Waste Management

- Monitor solid waste management practices
- Maintain current waste management planning

Liquid Waste Management

- Monitor liquid waste management practices
- Maintain current liquid waste management planning

Adequate Health Services

- Monitor health service accessibility
- Maintain current health facility planning

Assessment of The Healthy Environment

- Monitor environmental health conditions
- Maintain current environmental health planning

Assets and Utilities

Housing

- Monitor housing safety and climate impacts

- Maintain current housing safety planning

Public facilities

- Monitor public facility resilience and climate impacts

- Maintain current public facility planning

Storm Water Drainage

- Monitor storm water drainage functionality

- Maintain current drainage planning

Sanitary facilities

- Monitor sanitary facility adequacy

- Maintain current sanitation planning

Sewer lines

- Monitor sewer system functionality

- Maintain current sewer system planning

Power lines/supply

- Monitor power infrastructure resilience

- Maintain current power infrastructure planning

Energy for HH use

- Monitor household energy access

- Maintain current energy access planning

Water pipes

- Monitor water pipe network functionality

- Maintain current water infrastructure planning

Potable Water Treatment

- Monitor water treatment facility functionality

- Maintain current water treatment planning

Water Storage

- Monitor water storage facility safety

- Maintain current water storage planning

Roads & Transport

- Monitor road and transport infrastructure resilience

- Maintain current transport infrastructure planning

Communication

- Monitor communication infrastructure resilience

- Maintain current communication infrastructure planning

Potable water

Access

- Monitor potable water access

- Maintain current water access planning

Scarcities

- Monitor water scarcity patterns

- Maintain current water scarcity planning

Quality

- Monitor water quality

- Maintain current water quality planning

EBS

Soil

- Monitor soil conditions
- Maintain current soil management planning

Land Cover & Permeability

- Monitor land cover and permeability
- Maintain current land cover planning

Natural Vegetation

- Monitor vegetative cover
- Maintain current vegetation planning

Wetland

- Monitor wetland conditions
- Maintain current wetland protection planning

Biodiversity

- Monitor biodiversity conditions
- Maintain current biodiversity protection planning

Near Water Body

- Monitor proximity to water bodies
- Maintain current riparian protection planning

Sea

- Monitor sea level changes
- Maintain current coastal protection planning

Environmental Factors

- Monitor environmental degradation
- Maintain current environmental protection planning

Land use

- Monitor land use compliance
- Maintain current land use planning

Topography

- Monitor topography risks
- Maintain current topography planning

Institutions

Early Warning System

- Monitor early warning system functionality
- Maintain current early warning planning

Awareness & Information

- Monitor access to weather information
- Maintain current information access planning

Engagement & Communication

- Monitor community-government interaction
- Maintain current engagement planning

Representation & Reflection in Planning

- Monitor community representation in governance

- Maintain current representation planning

Health Facilities Operations

- Monitor health facility adequacy
- Maintain current health facility planning

Disaster Preparedness

- Monitor critical infrastructure identification
- Maintain current infrastructure protection planning

Disaster Response

- Monitor disaster response preparedness
- Maintain current disaster response planning

Post-Flood Recovery

- Monitor recovery finance availability
- Maintain current recovery finance planning

Public Facilities Maintenance

- Monitor infrastructure maintenance systems
- Maintain current maintenance planning

Design Recommendations

Livelihoods

Food Security

- Design climate-resilient food storage facilities with temperature control
- Design irrigation systems for food production
- Design protective structures for food crops

Grazing/Farm Land

- Maintain current grazing infrastructure design standards

Crops

- Maintain current crop protection infrastructure design standards

Livestock

- Maintain current livestock infrastructure design standards

Fishstock

- Maintain current fisheries infrastructure design standards

Local Commerce

- Maintain current commercial infrastructure design standards

Gender

- Maintain gender-responsive infrastructure design standards

Conviviality

- Maintain community-friendly infrastructure design

Education

- Maintain current school infrastructure design standards

Water For Irrigation / Water For Domestic Use

- Maintain current water infrastructure design standards

Access to Finance / Income

- Maintain current financial infrastructure design standards

Disaster Events

- Maintain current disaster-resilient infrastructure design standards

Health & Safety

Borne Diseases

- Maintain current health facility design standards

Safety, Location, Urban Planning & Conditions

- Maintain current housing design standards

Solid Waste Management

- Maintain current waste management infrastructure design standards

Liquid Waste Management

- Maintain current liquid waste infrastructure design standards

Adequate Health Services

- Maintain current health facility design standards

Assessment of The Healthy Environment

- Maintain current environmental protection design standards

Assets and Utilities

Housing

- Maintain current housing design standards

Public facilities

- Maintain current public facility design standards

Storm Water Drainage

- Maintain current drainage design standards

Sanitary facilities

- Maintain current sanitary facility design standards

Sewer lines

- Maintain current sewer system design standards

Power lines/supply

- Maintain current power infrastructure design standards

Energy for HH use

- Maintain current energy infrastructure design standards

Water pipes

- Maintain current water infrastructure design standards

Potable Water Treatment

- Maintain current water treatment facility design standards

Water Storage

- Maintain current water storage design standards

Roads & Transport

- Maintain current transport infrastructure design standards

Communication

- Maintain current communication infrastructure design standards

Potable water

Access

- Maintain current water access infrastructure design standards

Scarcities

- Maintain current water scarcity mitigation design standards

Quality

- Maintain current water quality treatment design standards

EBS

Soil

- Maintain current soil protection design standards

Land Cover & Permeability

- Maintain current permeability design standards

Natural Vegetation

- Maintain current vegetation design standards

Wetland

- Maintain current wetland protection design standards

Biodiversity

- Maintain current biodiversity protection design standards

Near Water Body

- Maintain current riparian protection design standards

Sea

- Maintain current coastal protection design standards

Environmental Factors

- Maintain current environmental protection design standards

Land use

- Maintain current land use design standards

Topography

- Maintain current topography design standards

Institutions

Early Warning System

- Maintain current early warning system design standards

Awareness & Information

- Maintain current information infrastructure design standards

Engagement & Communication

- Maintain current engagement infrastructure design standards

Representation & Reflection in Planning

- Maintain current representation infrastructure design standards

Health Facilities Operations

- Maintain current health facility design standards

Disaster Preparedness

- Maintain current infrastructure protection design standards

Disaster Response

- Maintain current disaster response infrastructure design standards

Post-Flood Recovery

- Maintain current recovery finance infrastructure design standards

Public Facilities Maintenance

- Maintain current maintenance infrastructure design standards

Community Development Plan Recommendations

Livelihoods

Food Security

- Develop emergency food distribution programs
- Create awareness programs on food security and climate impacts
- Facilitate access to climate-resilient crop varieties
- Develop programs for food production diversification and alternative food sources

Grazing/Farm Land

- Maintain current grazing land management practices
- Monitor grazing land quality and availability

Crops

- Maintain current crop production support programs
- Monitor crop yields and climate impacts

Livestock

- Maintain current livestock health programs
- Monitor livestock conditions and climate impacts

Fishstock

- Maintain current fisheries management programs
- Monitor fish stock levels and water quality

Local Commerce

- Maintain current business support programs
- Monitor business impacts from climate events

Gender

- Maintain gender-sensitive programs
- Monitor gender-differentiated climate impacts

Conviviality

- Maintain peace-building programs
- Monitor community relations and climate-related tensions

Education

- Maintain current education continuity programs
- Monitor school access during climate events

Water For Irrigation / Water For Domestic Use

- Maintain current water access programs
- Monitor water availability for non-drinking uses

Access to Finance / Income

- Maintain current financial support programs
- Monitor household financial impacts from climate events

Disaster Events

- Maintain current disaster preparedness programs
- Monitor disaster risks and community preparedness

Health & Safety

Borne Diseases

- Maintain current disease prevention programs
- Monitor disease incidence

Safety, Location, Urban Planning & Conditions

- Maintain current housing safety awareness programs

Solid Waste Management

- Maintain current waste management programs
- Monitor waste disposal practices

Liquid Waste Management

- Maintain current liquid waste management programs

Adequate Health Services

- Maintain current health service programs

Assessment of The Healthy Environment

- Maintain current environmental health programs

Assets and Utilities

Housing

- Maintain current housing safety programs

Public facilities

- Maintain current public facility management programs

Storm Water Drainage

- Maintain current drainage maintenance programs

Sanitary facilities

- Maintain current sanitation programs

Sewer lines

- Maintain current sewer system maintenance programs

Power lines/supply

- Maintain current power infrastructure maintenance programs

Energy for HH use

- Maintain current energy access programs

Water pipes

- Maintain current water infrastructure maintenance programs

Potable Water Treatment

- Maintain current water treatment programs

Water Storage

- Maintain current water storage programs

Roads & Transport

- Maintain current transport infrastructure maintenance programs

Communication

- Maintain current communication infrastructure maintenance programs

Potable water

Access

- Maintain current water access programs

Scarcities

- Maintain current water scarcity programs

Quality

- Maintain current water quality programs

EBS

Soil

- Maintain current soil management programs

Land Cover & Permeability

- Maintain current land cover programs

Natural Vegetation

- Maintain current vegetation programs

Wetland

- Maintain current wetland protection programs

Biodiversity

- Maintain current biodiversity protection programs

Near Water Body

- Maintain current riparian protection programs

Sea

- Maintain current coastal protection programs

Environmental Factors

- Maintain current environmental protection programs

Land use

- Maintain current land use programs

Topography

- Maintain current topography programs

Institutions

Early Warning System

- Maintain current early warning programs

Awareness & Information

- Maintain current information access programs

Engagement & Communication

- Maintain current engagement programs

Representation & Reflection in Planning

- Maintain current representation programs

Health Facilities Operations

- Maintain current health facility programs

Disaster Preparedness

- Maintain current infrastructure protection programs

Disaster Response

- Maintain current disaster response programs

Post-Flood Recovery

- Maintain current recovery finance programs

Public Facilities Maintenance

- Maintain current maintenance programs